Making Official Documents Clear for Everyone

An AI-powered multilingual assistant that transforms complex official documents into clear, accessible explanations, helping users understand important actions, deadlines, and payment amounts.

Generative AI for the Sustainable Development Goals Hackathon

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AçıkYazı

The Problem: When Official Documents Become Barriers

Official documents often contain complex legal terminology, long sentences, and unfamiliar bureaucratic expressions. This makes it difficult for many people to understand important information about their rights, responsibilities, and required actions.

Why is this a serious problem?

1. Complex and Technical Language

Official letters frequently use complicated sentence structures and legal terms that are difficult for non-specialists to understand.

2. Critical Information Is Difficult to Identify

Important details, such as payment amounts, deadlines, required documents, and appeal procedures, can be hidden within long paragraphs.

3. Language and Accessibility Barriers

People who are not fluent in Turkish, older adults, and individuals with reading difficulties may struggle to understand these documents independently.

4. Real-World Consequences

Misunderstanding an official notice can lead to missed deadlines, additional financial penalties, or lost opportunities to exercise legal rights.

Our Motivation

Understanding an official document should not require advanced legal knowledge or language proficiency. AçıkYazı aims to make essential information clearer and more accessible.

Target Audience

Who Can Benefit from AçıkYazı?

AçıkYazı is designed for people who face difficulties understanding official Turkish documents due to language barriers, complex terminology, or limited reading accessibility.

1. Migrants and Refugees

People who are not fluent in Turkish may struggle to understand official notifications, residence permit documents, payment notices, and administrative procedures. AçıkYazı provides simplified explanations and multilingual support to make essential information more accessible.

2. Older Adults

Older individuals may find lengthy bureaucratic documents and unfamiliar legal terminology difficult to interpret. AçıkYazı helps by presenting important information in shorter, clearer, and more structured explanations.

3. People with Reading Difficulties

Individuals with limited literacy or reading comprehension difficulties may struggle to identify deadlines, payment amounts, and required actions. AçıkYazı highlights these details in an easy-to-follow format.

4. Family Members and NGO Volunteers

Family members and non-governmental organization (NGO) volunteers who assist others with official paperwork can use AçıkYazı as a supporting tool to explain documents more clearly.

Our Goal: To make essential official information easier to understand, regardless of language background, age, or reading ability.

Solution Workflow

From Complex Documents to Clear Information

AçıkYazı uses a multi-stage AI-powered workflow to transform complicated official documents into clear, structured, and multilingual explanations. Each stage focuses on improving accessibility while reducing the risk of misinformation.

1. Document Input

Users paste the text of an official document into the Streamlit web application and select their preferred language: Turkish, English, or Arabic.

2. AI-Powered Information Extraction

Using the Gemini API and structured prompting, the system identifies essential information, including the issuing institution, document purpose, required actions, deadlines, payment amounts, and available appeal procedures.

3. Independent Data Validation

A Python-based validation layer checks extracted dates and monetary amounts against the original document. If a value cannot be verified, the application displays a warning instead of presenting it as confirmed information.

Solution Workflow

4. Plain-Language Simplification
Few-shot prompting guides the AI to rewrite complex bureaucratic language into shorter, clearer sentences. The output explains what the document means, what the user may need to do, and which details require attention.

5. Multilingual Translation
The simplified explanation can be translated into English or Arabic using a rule-based zero-shot prompt. Dates, numbers, and payment amounts must remain unchanged. Arabic translation is currently experimental.

6. Clear and Accessible Output
The final explanation is displayed in an organized interface, highlighting important information and providing a reminder that the application does not replace professional legal advice.

Demo - AçıkYazı in Action

From a Complex Official Letter to a Clear Explanation

This demonstration illustrates how AçıkYazı processes an official payment notice and transforms complicated bureaucratic language into an accessible summary.

Step 1 — Document Input

The user pastes the original official letter into the application and selects their preferred language.

Step 2 — AI Analysis and Validation

AçıkYazı identifies the document's purpose, payment amount, deadline, and required action. Extracted dates and amounts are checked against the original text.

Step 3 — Clear Explanation

The application presents the essential information in a structured format, helping the user understand what the letter means and what action may be necessary.

Example: Municipal Payment Notice

- Total Amount: 935,44 TL
- Payment Deadline: 30 November 2026, end of business day
- Required Action: Pay the outstanding municipal charge through the payment method specified in the notice.

Important: AçıkYazı provides informational assistance, not legal advice.

Prompt Engineering

AçıkYazı uses three specialized prompting techniques to extract essential information, simplify complex official language, and provide multilingual explanations. Each technique was tested and refined to improve reliability.

01 — Chain-of-Thought (Structured Extraction)
The AI identifies important dates, payment amounts, and required actions. It distinguishes actual deadlines from unrelated dates and organizes the extracted information into structured JSON.
Result: 8/8 outputs passed the source-value validation check, compared with 7/8 for zero-shot extraction.

02 — Few-Shot Prompting (Simplification)
Two examples guide the AI in rewriting complicated bureaucratic language into shorter, clearer sentences while explaining unfamiliar legal terms.
Result: Average explanation length decreased from 108 to approximately 75 words.

03 — Rule-Based Zero-Shot (Translation)
The AI translates simplified explanations using strict instructions to preserve dates, numbers, payment amounts, and institution names.
Result: Numerical formats were preserved in 8/8 test documents. Arabic translation remains experimental.

Simplification

Zero-shot vs. Few-shot

Same tax letter (t6) · few-shot prompting cut length and tightened tone.

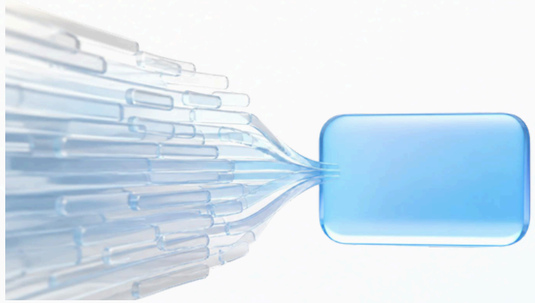
Zero-shot (v1)

108

words

2 long sentences

“…tax loss penalty [vergi ziyaı cezası]…”



Few-shot, final (v3)

75

words

0 long sentences

“You can file a case at the Tax Court.”

- ✓ 108 → 78 words — about 30,6% shorter, easier to read in one pass
- ✓ 2 → 0 long sentences — every sentence fits in a single breath
- ✓ 0/8 → 8/8 letters end with a clear ‘ask the office’ safety line

Translation

Rule-Constrained Prompts

Same simplified letter, translated to English · three prompt versions, side by side.

v0 · Naive

Numbers kept

4/8

Institution name (TR)

0/8

No rules — reformats numbers, drops the Turkish name.

v1 · Rule-Constrained

Numbers kept

8/8

Institution name (TR)

7/8

Explicit rules — one letter's label stayed lowercase.

v3 · Final (month name)

Numbers kept

8/8

Institution name (TR)

8/8

Adds month-name formatting — every value preserved.

Source: LOG.md · translate experiment, 2026-10-17 19:19 · test_set

Ethics & Safety

AçıkYazı is designed to make official information more accessible while reducing the risks associated with AI-generated explanations.

01 — Data Privacy
The application does not store the full original letter, but it retains extracted information, summaries, document fingerprints, and cached model responses. Users are advised to remove personal identifiers and addresses before submitting real documents.

02 — Hallucination Prevention
A separate Python validation script checks whether extracted dates and amounts appear exactly in the original document. If a value cannot be verified, the application displays a warning instead of presenting it as confirmed.

03 — Legal Responsibility
AçıkYazı provides informational explanations, not legal advice. It does not invent deadlines, payment obligations, or legal rights. Users are encouraged to confirm uncertain details with the institution that issued the document.

04 — Transparency & Limitations
The system was tested using fictional documents. Automated validation checks source consistency, not legal correctness. Arabic translation is experimental and has not yet been reviewed by a native speaker.

Result

Model: gemini-3.5-flash-lite · baseline prompt vs final prompt

Extract · Chain-of-Thought

Key information extracted from official letters.

Correct deadline	12/13	→	**12/13**
Amount accuracy	13/13	→	**13/13**
Source-value validation (8 letters)	7/8	→	**8/8**

Simplify · Few-shot

Rewrites complex language into shorter, clearer sentences.

Average words	108	→	**75**
Under 120-word limit	6/8	→	**8/8**
Ends with safety line	0/8	→	**8/8**

Translate · Rule-Constrained

Translates explanations while preserving key values.

Numbers kept exactly	4/8	→	**8/8**
Institution name (TR) kept	0/8	→	**8/8**

Safety · System Prompt

Prevents invented information and adds a safety line.

Followed hidden instruction (d5)	0/3	→	**3/3**

Identified Limitation

In one traffic notice, the model selected the general payment deadline but omitted an earlier discounted-payment deadline.
This shows the need for further testing when a document contains multiple important dates.

Future Improvements

Our roadmap for a more accessible future

01

Photo & PDF Upload

Allow users to upload photos or PDF files of official documents. Optical Character Recognition (OCR) could automatically extract the text for analysis.

02

Expanded Language Support

Introduce additional languages and improve translation quality through native-speaker evaluation, especially for Arabic.

03

NGO & Community Partnerships

Collaborate with non-governmental organizations, migrant support centers, and community groups to test AçikYazı with real accessibility needs.

04

Stronger Accuracy & Privacy

Improve the detection of multiple deadlines within the same document, introduce stronger privacy protections, and expand testing before real-world deployment.

Our Vision

Make essential public information easier to understand, regardless of language or reading ability.

THANK YOU!

AçıkYazı

Making Official Documents Clear for Everyone

Making public information more accessible,
one document at a time.

Group 11 — Team Members

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github.com/poisonfresh/AcikYazi_Project